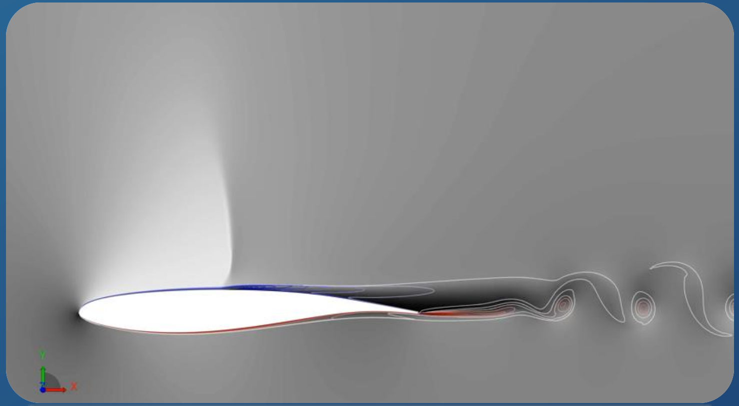
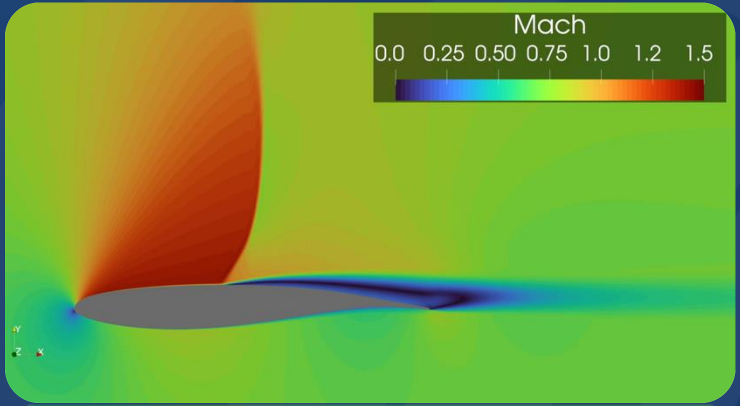
DPW-8/AePW-4: Transonic Shock Buffet URANS and LBM/VLES Simulations of the ONERA OAT15A Airfoil





Hadar Ben-Gida

Technion Israel Institute of Technology, Israel

21-25 July 2025, Las Vegas, NV

TECHNION Israel Institute of Technology

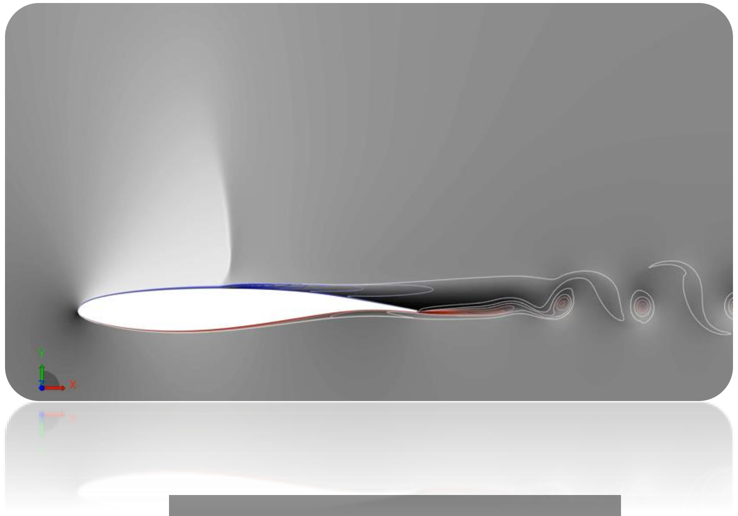
AVIATION FORUM

Introduction

Transonic shock buffet (type II)

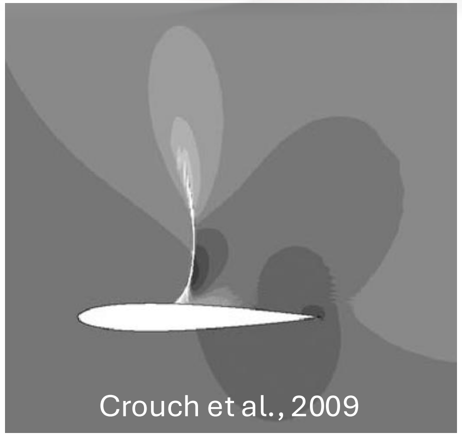
❑ Key features

- Self-sustained shock wave oscillations due to *interference* with a separated turbulent BL on the upper surface of a supercritical airfoil at $\alpha > 0$
- *Sensitive* to airfoil shape, Mach, and α
- *Low-frequency lift oscillations*, typically 60-100 Hz for WT scale [1-3]
- Poses *significant challenges* to transonic aircraft design and operation
    - Induced structural vibrations can deteriorate the performance and fatigue life



❑ Physical mechanisms on 2D airfoils

- *Lee's acoustic feedback loop model* [4]
    - UTW generated at the TE $\Leftrightarrow$ DTW generated inside the BL by the shock wave
- *Crouch's instability model* [5]
    - 2D buffet is a global flow instability, yielding phase-locked motion of the shock and separated BL



Crouch et al., 2009

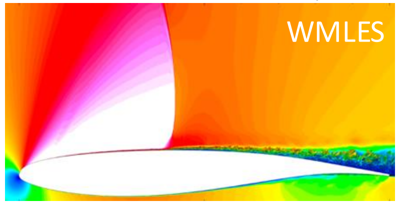
[1] Jacquin L. et al. (2009) "Experimental study of shock oscillation over a transonic supercritical profiles." AIAA Journal 47(9), p. 1985-1994.
[2] Deck S. (2005) "Numerical simulation of transonic buffet over a supercritical airfoil." AIAA Journal 43(7), p. 1556-1566.
[3] D'Aguanno A. et al. (2021) "Experimental investigation of the transonic buffet cycle on a supercritical airfoil." Experiments in Fluids 62(10), p. 214.
[4] Lee, B. H. K. (1990) "Oscillatory shock motion caused by transonic shock boundary-layer interaction." AIAA Journal 28(5), p. 942-944
[5] Crouch J. D. et al. (2009) "Origin of transonic buffet on aerofoils." Journal of Fluid Mechanics 628, p. 357-369.

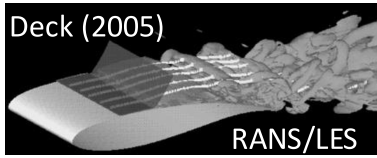
Introduction

2D airfoil transonic shock buffet – CFD predictions

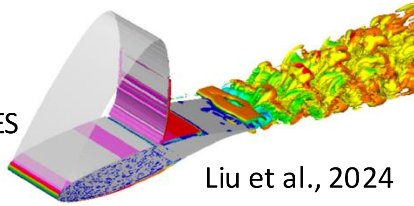
- **Previous studies on 2D transonic shock buffet employed various modeling approaches**
  - URANS [6,12], RANS/LES [2,7,8], WMLES [9,10], LBM/VLES [11]
  - *Varying degrees of accuracy*
- **DPW-8/AePW-4 Buffet Working Group**
  - Evaluating the effectiveness of existing CFD tools and methods
  - Determine best practices to accurately resolve unsteady, fixed-geometry at buffet conditions

Fukushima et al., 2018

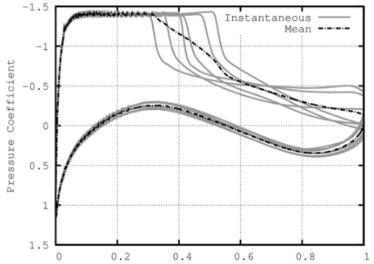




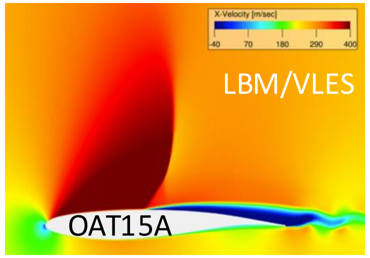
RANS/LES

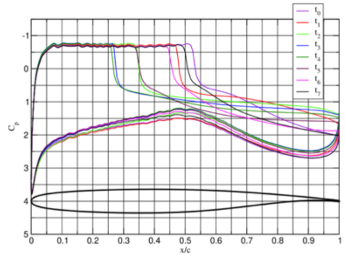


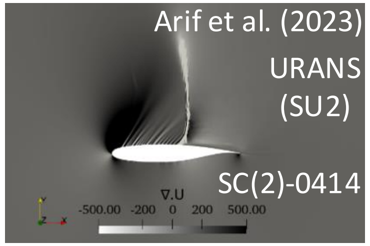
Liu et al., 2024



Ribeiro et al. (2017)



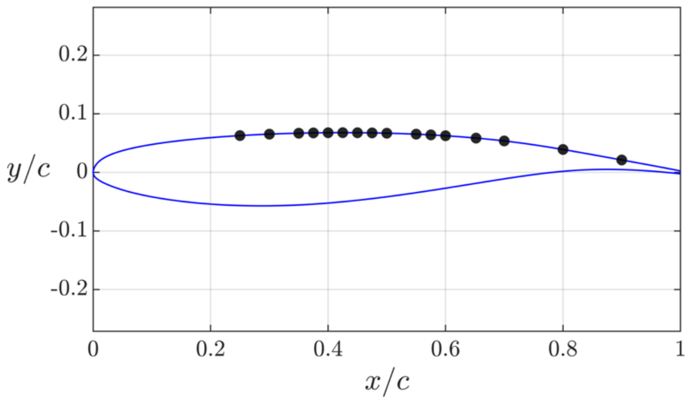


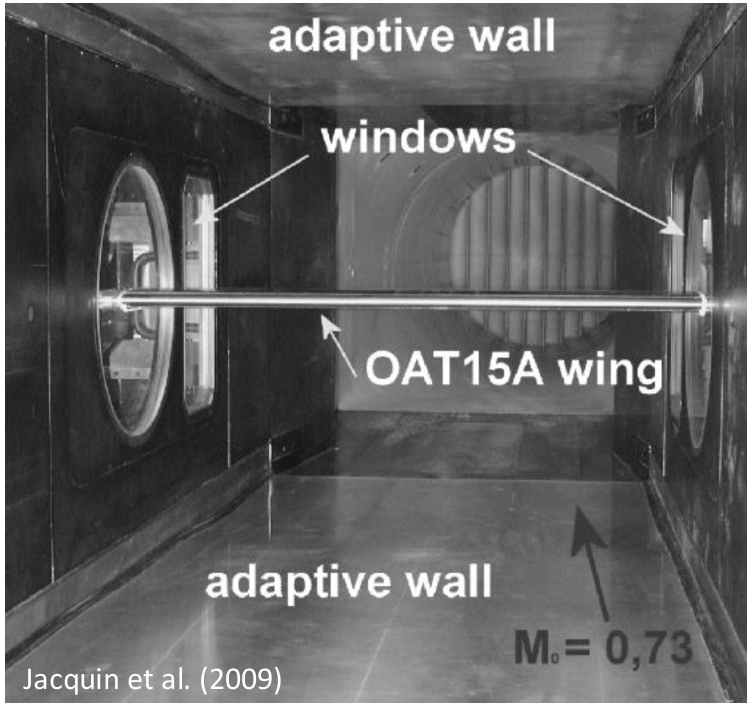


[6] Poplingher L. and Raveh D. E. (2023) "Comparative modal study of the two-dimensional and three-dimensional transonic shock buffet." AIAA Journal 61(1), p. 125-144.
[7] Grossi F. et al. (2014) "Prediction of transonic buffet by delayed detached-eddy simulation." AIAA Journal 52(10), p. 2300-2312.
[8] Liu T. et al. (2024) "Prediction of transonic shock buffet over supercritical airfoil OAT15A based on zonal detached-eddy simulation." Applied Sciences 14(21), p. 9628.
[9] Fukushima Y. and Kawai S. (2018) "Wall-modeled large-eddy simulation of transonic airfoil buffet at high reynolds number." AIAA Journal 56(6), p. 2372-2388.
[10] Hass R. et al. (2024) "Wall-modeled large eddy simulations of transonic buffet over a supercritical airfoil." AIAA Aviation Forum, 29 July - 2 August, Las Vegas, NV, USA.
[11] Ribeiro, A. F. P. et al. (2017) "Buffet simulations with a lattice-Boltzmann based transonic solver." 55th AIAA Aerospace Sciences Meeting, 9-13 January, Grapevine, TX.
[12] Arif, M. et al. (2023) "Two-dimensional transonic buffet in a supercritical wing section." AIAA SciTech Forum, 23-27 January, National Harbor, MD & Online.

DPW-8/AePW-4 Buffet Working Group
Test Case 1

- **Test Case 1** - Experimental data from Jacquin et al. (2009)
  - Continuous closed-circuit transonic S3Ch WT (ONERA-Meudon Center)
  - *ONERA OAT15A* supercritical airfoil
  - $c = 230$ mm, $b = 780$ mm, $AR = 3.4$ (blunt TE)
  - Tripped @ $x = 0.07c$
  - 36 *unsteady* pressure transducers (●)
  - $M_\infty = 0.73$, $Re_c = 3 \times 10^6$
  - $T_\infty = 271$ K, $p_\infty = 71.8$ kPa, $p_0 = 102.4$ kPa
  - $\alpha = 1.36° \longrightarrow 3.90°$ (buffet onset @ $\alpha = 3.10°$)
  - Estimated freestream turbulence intensity value is $TI \approx 0.26\%$ [13]
- **Test Case 1a –** *Steady* RANS
- **Test Case 1b –** *Unsteady* CFD

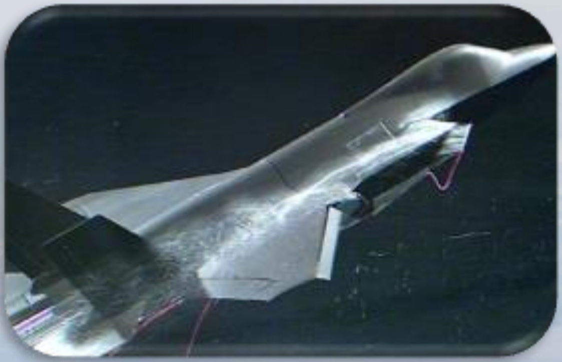




Jacquin et al. (2009)

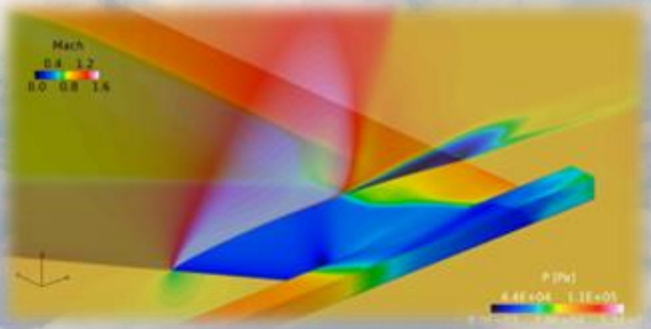
[1] Jacquin L. et al. (2009) "Experimental study of shock oscillation over a transonic supercritical profiles." AIAA Journal 47(9), p. 1985-1994.
[13] Brion V. et al. (2020) "Laminar buffet and flow control." Proceedings of the Institution of Mechanical Engineers, Part G: Journal of Aerospace Engineering 234(1), p. 124-139.

Objectives of the Present Study



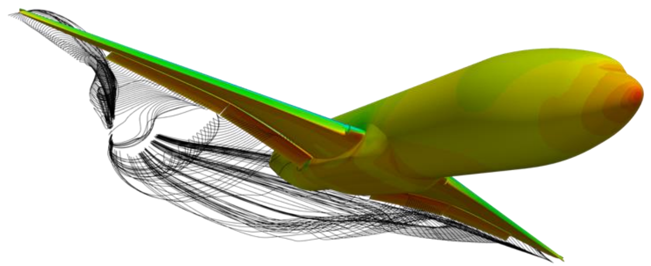
- **Compare the *accuracy*** in predicting the 2D shock buffet phenomenon on the OAT15A between ***two different CFD approaches***:
  - **URANS (SU2)**
  - **LBM/VLES (PowerFLOW)**

- **Evaluate the *performance* against WTT data:**
  - Aerodynamic polars
  - Mean and rms Cp distributions
  - Spectral content



- Develop ***best practices*** to be applied on **realistic transonic aircraft**

Numerical Methodology
URANS



- SU2 code v8.1.0
- Flow field was initialized with steady RANS, followed by time-accurate URANS simulations
- **Core numerical scheme**:
  - ***Convective flux***: Jameson-Schmidt-Turkel (JST) [14] with the Venkatakrishnan slope limiter [15]
  - ***Turbulence model***: SA-EW-CC-QCR
  - ***Dual-time-stepping strategy***: 50 inner iterations, $\Delta t = 2.5 \cdot 10^{-6}$ sec
  - ***Linear solver***: Generalized minimal residual (GMRES)
  - ***Convergence***: RANS – 10^{-7}; URANS – 10^{-6} per time step
  - ***Simulation time***: 1,050 CTUs (1 sec)

SA - Spalart-Allmaras
EW - Edwards and Chandra's modification [16]
CC- compressibility correction [17]
QCR - quadratic constitutive relation (v2000) [18]

- **Sensitivity analysis** was conducted to study effects of:
  - Upwind-based convective flux schemes, low dissipation JST coefficients, slope limiters, turbulence model variants, and physical time step

[14] Jameson A., et al. (1981) "Numerical solution of the Euler equations by finite volume methods using Runge Kutta time stepping schemes." 14th Fluid and Plasma Dynamics Conference, June 23-25, Palo Alto, CA, USA.
[15] Venkatakrishnan V. (1995) "Convergence to steady state solutions of the Euler equations on unstructured grids with limiters." Journal of Computational Physics 118(1), p. 120-130.
[16] Edwards J. R. and Chandra S. (1996) "Comparison of eddy viscosity-transport turbulence models for three-dimensional, shock-separated flow fields." AIAA Journal 34(4) p. 756-763.
[17] Spalart P. (2000) "Trends in turbulence treatments." Fluids 2000 Conference and Exhibit, 19-22 June, Denver, CO, USA.
[18] Spalart P. (2000) "Strategies for turbulence modelling and simulations." International Journal of Heat and Fluid Flow 21(3), p. 252-263.

Numerical Methodology
LBM/VLES

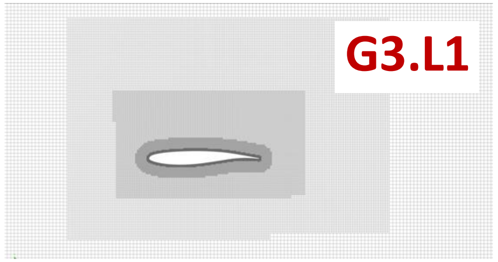
- SIMULIA PowerFLOW (v6-2025R1)
- **Core scheme**:
  - ***Transonic flow solver*** (39-state LBM) [19]
  - *Bhatnagar-Gross-Krook* (**BGK**) ***collision model*** [20]
  - ***RNG k-ϵ turbulence model***
    - Recalibrate the collision model to characteristic turbulent flow time scales [21,22]
    - Eddy viscosity ratio was set to $\nu_t/\nu = 26$ ($l_t = 1.14$ mm)
  - ***Wall function***
    - With extension of the standard "law of the wall" to account for the effects of favorable and adverse pressure gradients [23]
  - **Variable cubic Cartesian grid + immersed boundary method**
  - $CFL < 1$
  - ***Simulation time***: 680 CTUs (0.65 sec)

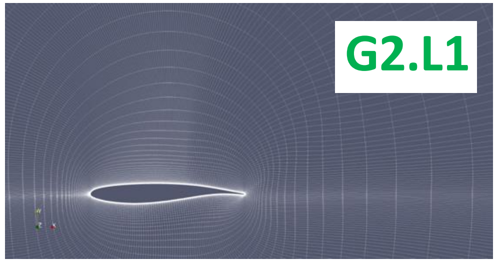
$$\hat{f}_i(\vec{x} + c_i\Delta t, t + \Delta t) = \hat{f}_i(\vec{x}, t) + C_i(\vec{x}, t),$$

$$C_i(\vec{x}, t) = -\left[\hat{f}_i(\vec{x}, t) - \hat{f}_i^{\mathrm{eq}}(\vec{x}, t)\right]\frac{\Delta t}{\tau}$$

[19] Nie X. et al. (2009) "A lattice-Boltzmann / finite-difference hybrid simulation of transonic flow." 47th AIAA Aerospace Sciences Meeting, 5-8 January, Orlando, FL.
[20] Bhatnagar et al. (1954) "A model for collision processes in gases. I. Small amplitude processes in charged and neutral one-component systems." Physical Review 94, p. 511-525
[21] Chen, H. et al. (2003) "Extended-Boltzmann kinetic equation for turbulent flows." Science 5633, p. 633-636.
[22] Yakhot V. and Orszag S. A. (1986) "Renormalization group analysis of turbulence. I. Basic theory." In: Journal of Scientific Computing 1(1), p. 3-51.
[23] Fares E. and Noelting S. (2011) "Unsteady flow simulation of a high-lift configuration using a lattice Boltzmann approach." 49th AIAA Aerospace Sciences Meeting, 4-7 January, Orlando, FL.

Grid Configurations
Overview





- **URANS**
  - *Unstructured* (G1)
    - Cadence-supplied grids
    - 6 refinement levels
  - *Structured* (G2)
    - O-type custom grids [6]
    - 2 refinement levels
- **LBM/VLES**
  - *Cartesian* (G3)
    - 10 variable resolution (VR) regions
    - 5 refinement levels

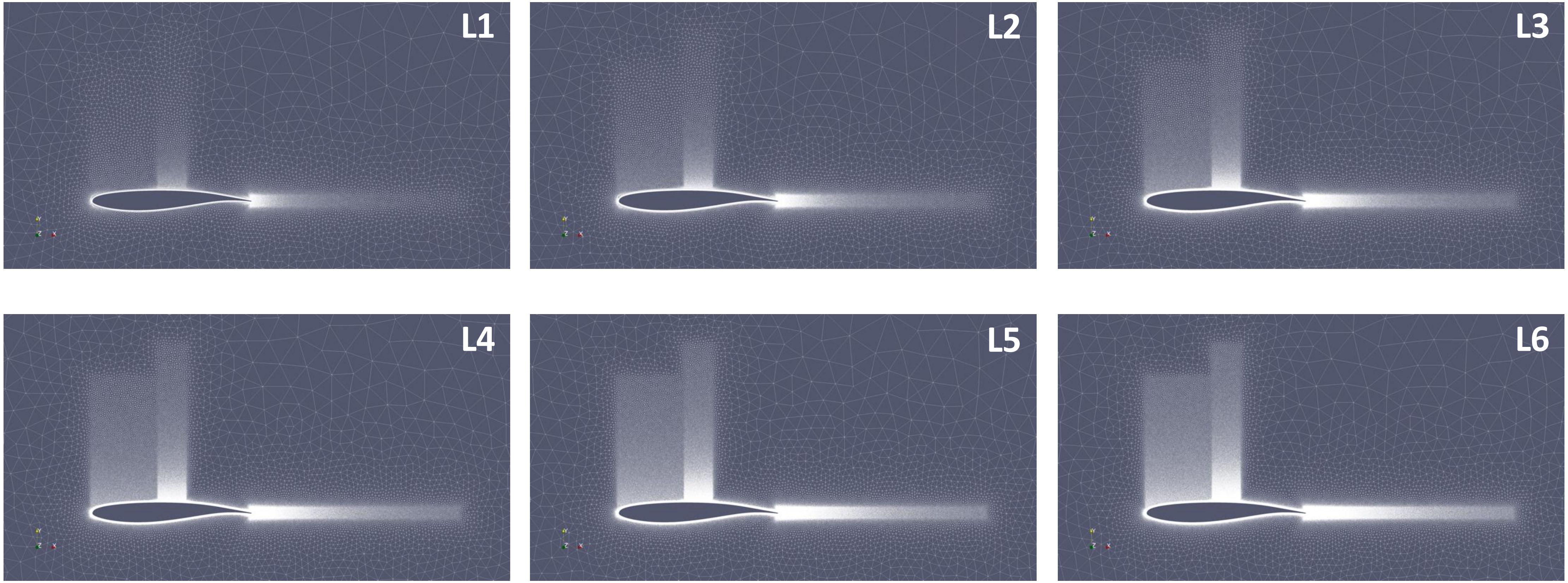
Grid family	ID	Refinement level	$\Delta y_w/c \times 10^6$	Far field extent	Total cell count	y^+
Cadence unstructured grid	G1.L1	1	6.85	152c	47,187	0.45
	G1.L2	2	4.57	152c	89,616	0.29
	G1.L3	3	3.41	152c	150,333	0.22
	G1.L4	4	2.74	152c	235,491	0.18
	G1.L5	5	2.28	152c	353,725	0.15
	G1.L6	6	1.96	152c	517,448	0.13
Custom structured O-type grid	G2.L1	1	4.35	36c	48,360	0.29
	G2.L2	2	2	218c	179,200	0.13
PowerFLOW Cartesian grid	G3.L1	1	2,000	60c	105,703	120
	G3.L2	2	1,000	60c	404,244	55
	G3.L3	3	665	60c	896,131	35
	G3.L4	4	570	60c	1,257,927	29
	G3.L5	5	500	60c	1,581,336	25

[6] Poplingher L. and Raveh D. E. (2023) "Comparative modal study of the two-dimensional and three-dimensional transonic shock buffet." AIAA Journal 61(1), p. 125-144.

Grid Configurations
Unstructured (G1)

Grid family	ID	Refinement level	$\Delta y_w/c$ $\times 10^6$	Far field extent	Total cell count	y^+
Cadence unstructured grid	G1.L1	1	6.85	$152c$	47,187	0.45
	G1.L2	2	4.57	$152c$	89,616	0.29
	G1.L3	3	3.41	$152c$	150,333	0.22
	G1.L4	4	2.74	$152c$	235,491	0.18
	G1.L5	5	2.28	$152c$	353,725	0.15
	G1.L6	6	1.96	$152c$	517,448	0.13

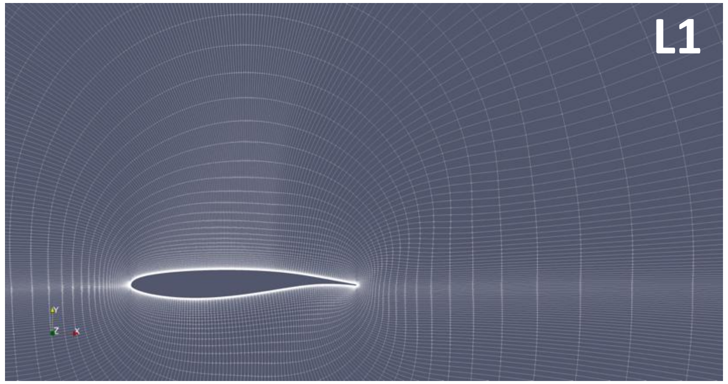
- ❑ Used for SU2

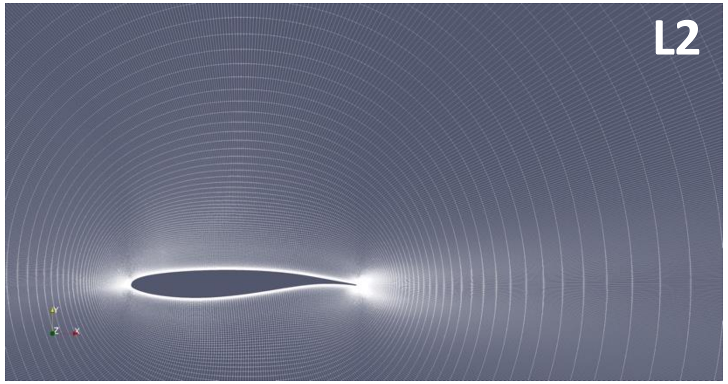


Grid Configurations
Structured (G2)

Grid family	ID	Refinement level	$\Delta y_w/c$ $\times 10^6$	Far field extent	Total cell count	y^+
Custom structured O-type grid	G2.L1	1	4.35	$36c$	48,360	0.29
	G2.L2	2	2	$218c$	179,200	0.13

- ❑ Used for SU2

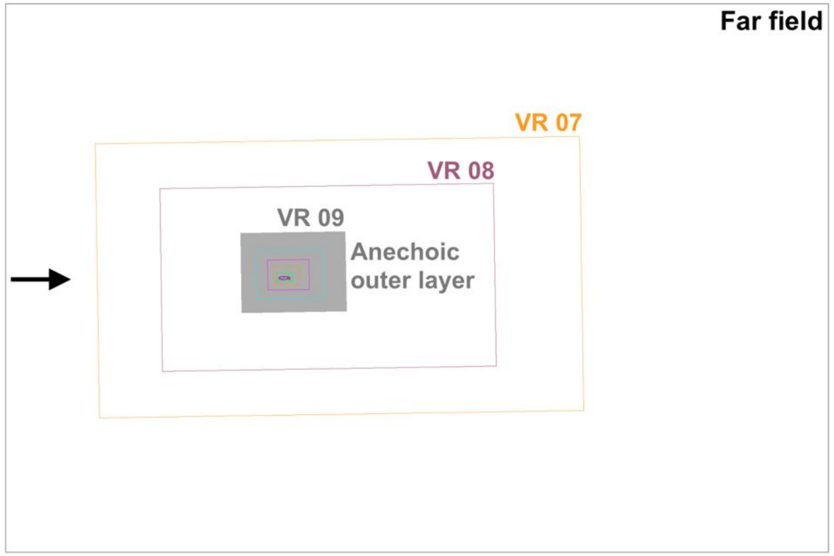


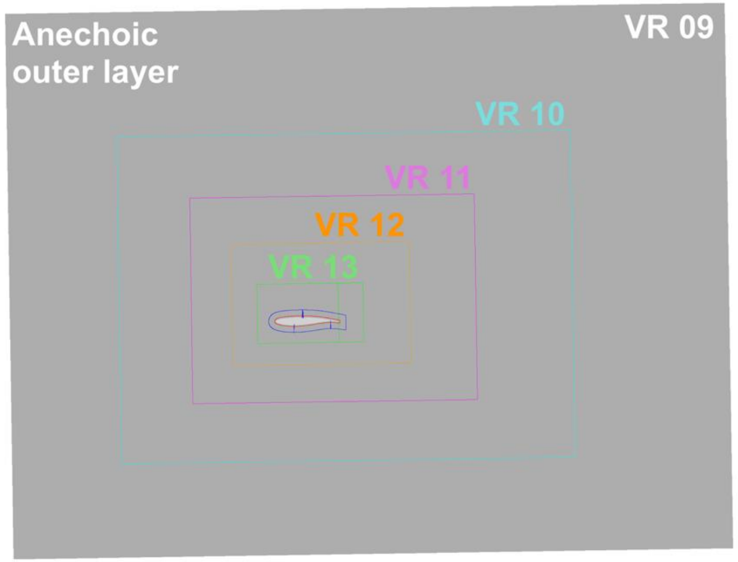


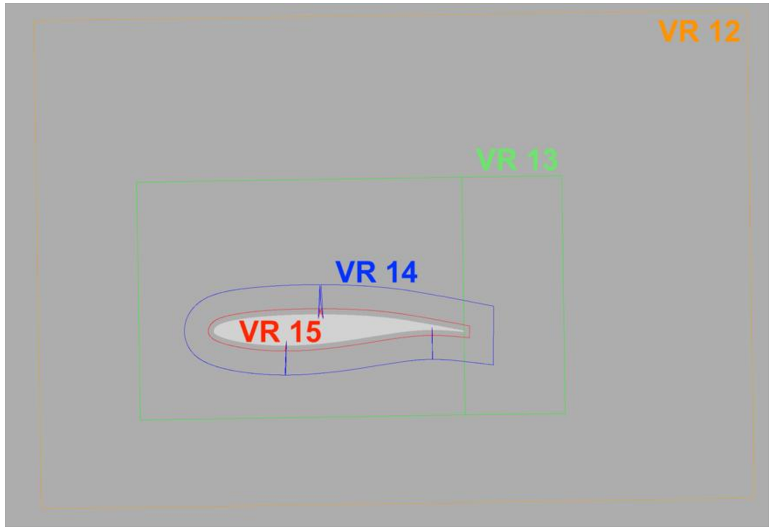
Grid Configurations
Cartesian (G3)

- Used for PowerFLOW

Grid family	ID	Refinement level	$\Delta y_w/c$ $\times 10^6$	Far field extent	Total cell count	y^+
PowerFLOW Cartesian grid	G3.L1	1	2,000	$60c$	105,703	120
	G3.L2	2	1,000	$60c$	404,244	55
	G3.L3	3	665	$60c$	896,131	35
	G3.L4	4	570	$60c$	1,257,927	29
	G3.L5	5	500	$60c$	1,581,336	25



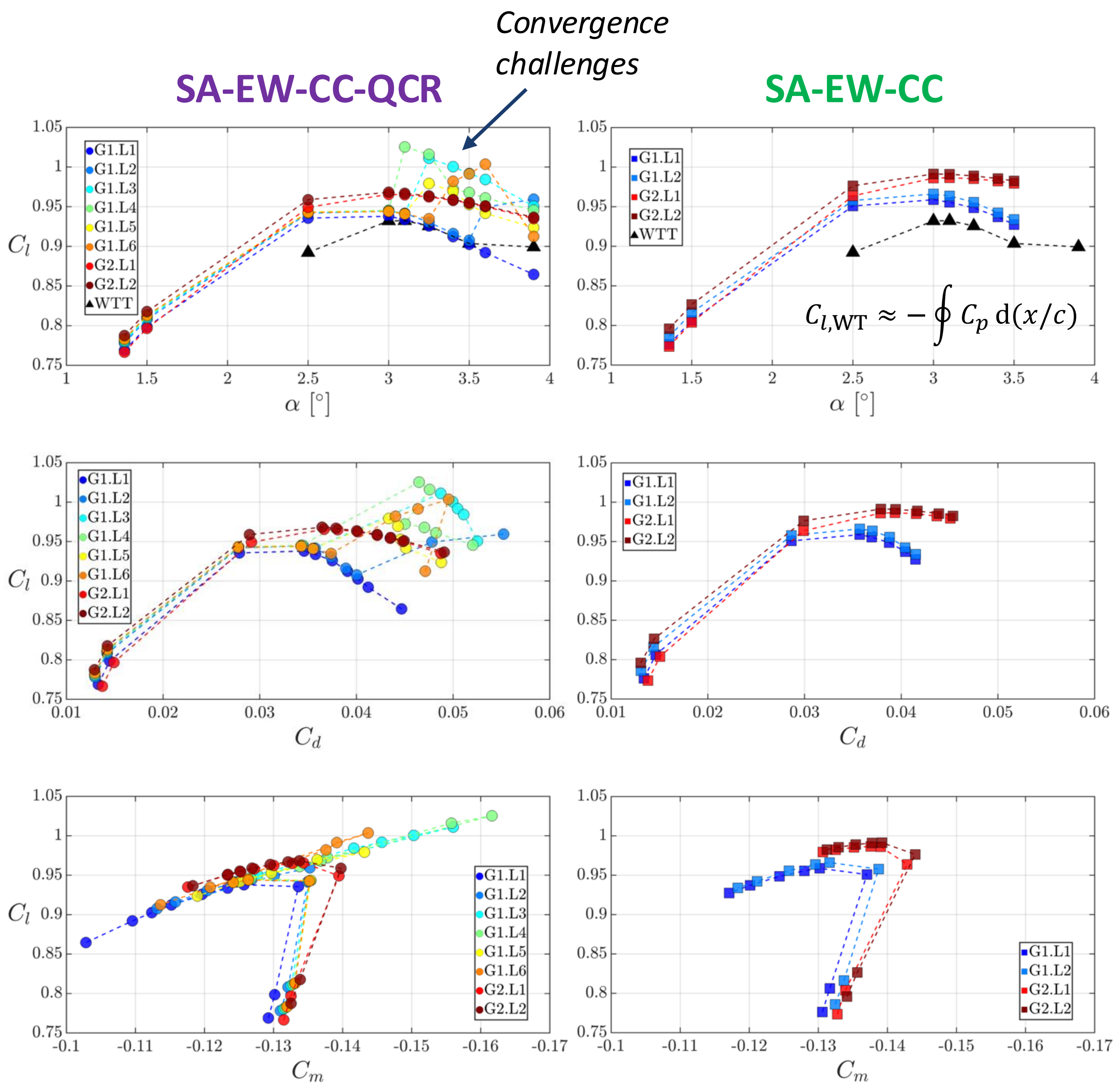




Results – Test Case 1a

Integrated F&M (RANS only)

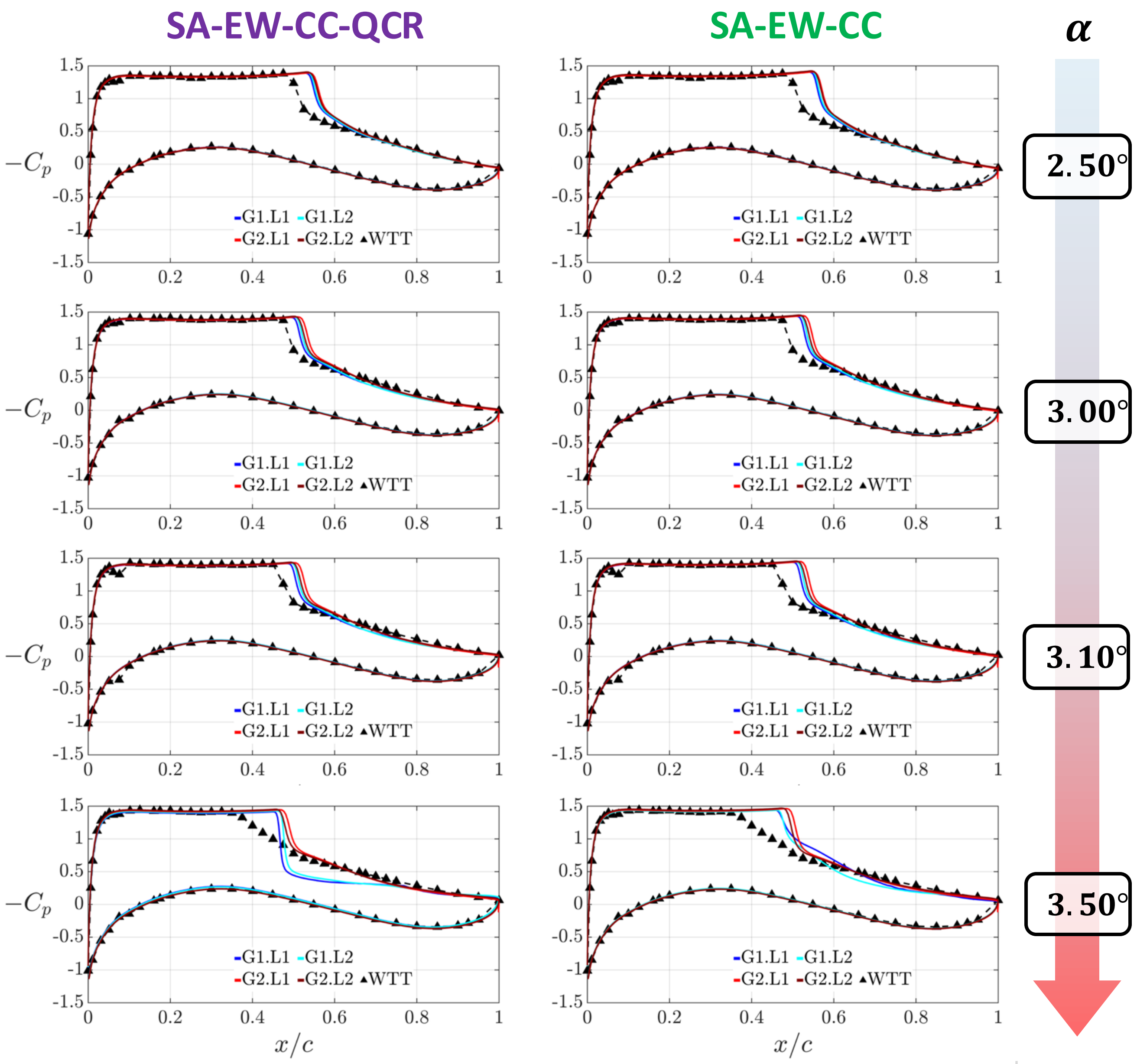
- ❑ ***With and without*** **the QCR correction**
- ❑ **Convergence**
  - ➢ G1.L3-L6 grids had *limited success* in reaching the convergence criteria (10^{-7}) at post-buffet onset conditions ($\alpha > 3.10°$) - *due to unsteady flow* (?)
  - ➢ QCR requires *more iterations* to converge
- ❑ **Lift**
  - ➢ G1.L1-L2 yielded closest C_l polars to the WTT data
  - ➢ G2 grids resulted in slightly higher C_l polars
  - ➢ QCR improves the C_l polar results
- ❑ **Drag**
  - ➢ *Low-α scatter*: 10 drag counts, $\Delta C_l \approx 0.02$
  - ➢ *High-α scatter*: 100 drag counts, $\Delta C_l \approx 0.17$
  - ➢ QCR results in higher drag at $\alpha > 3.10°$



Results – Test Case 1a

C_p distributions (RANS only)

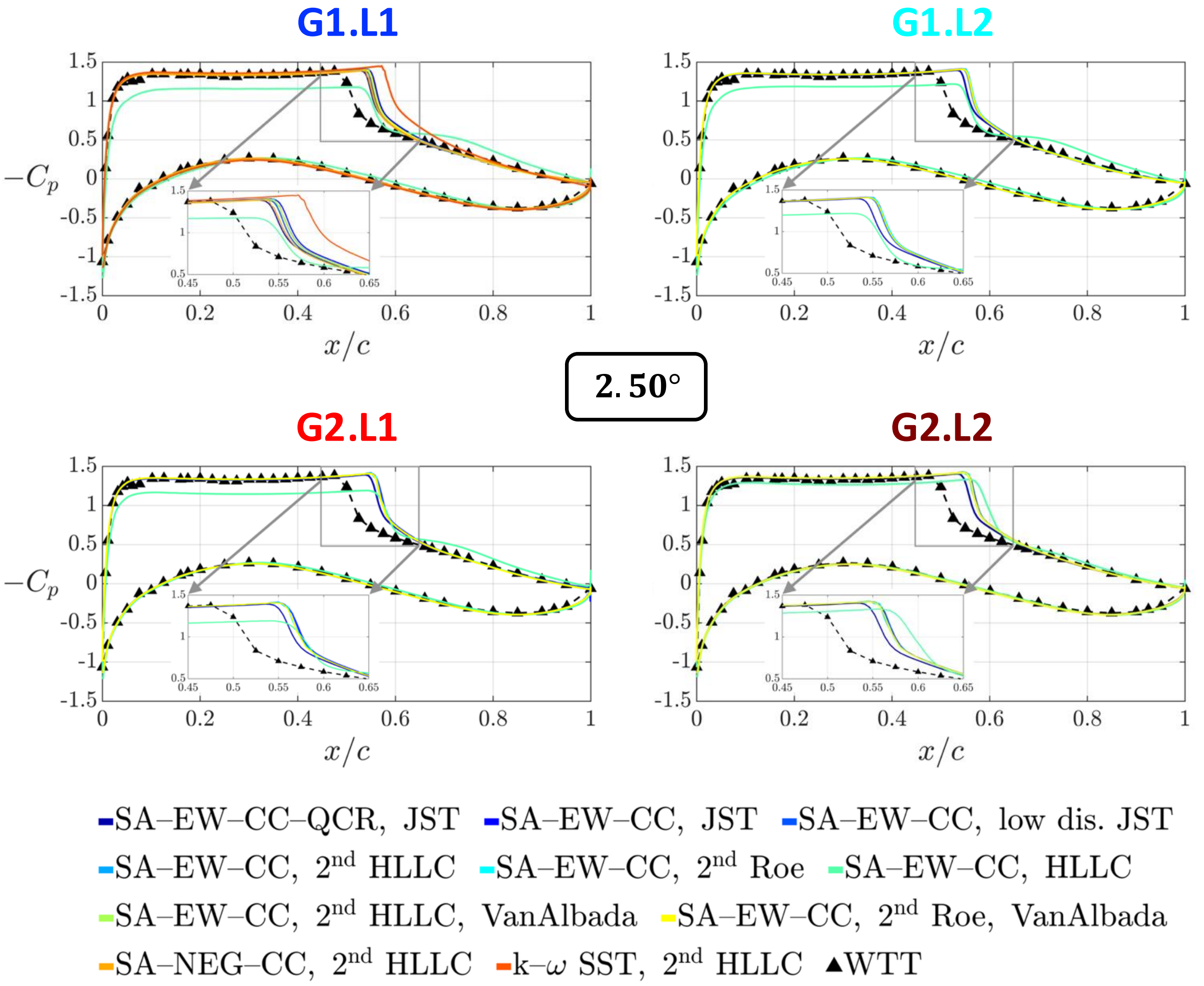
- **RANS results for G1/2.L1/2**
  - *w & w/o* QCR correction for various α
- **Shock** is captured *downstream* of WTT data
  - QCR shows some *improvement*
  - G1 shows slight improvement at $\alpha \leq 3.10°$
- At $\alpha \geq 3.10°$, RANS is unable to predict the C_p trend around the shock



Results – Test Case 1a

C_p distributions (RANS only)

- **Sensitivity analysis @ $\alpha = 2.50°$**
  - Convective flux schemes (JST, HLLC, Roe)
  - Limiters (Venkatakrishnan, van Albada)
  - Turbulence models (SA variants, k-ω SST)
- **Upwind schemes** with 2nd order reconstruction (MUSCL) yield *no significant* advantage over the JST
  - Same for reducing the 2nd and 4th order JST dissipation coefficients
- **Slope limiters** have a *negligible* effect
- ***k*-*ω* SST** captures the shock further downstream than any of the SA variants



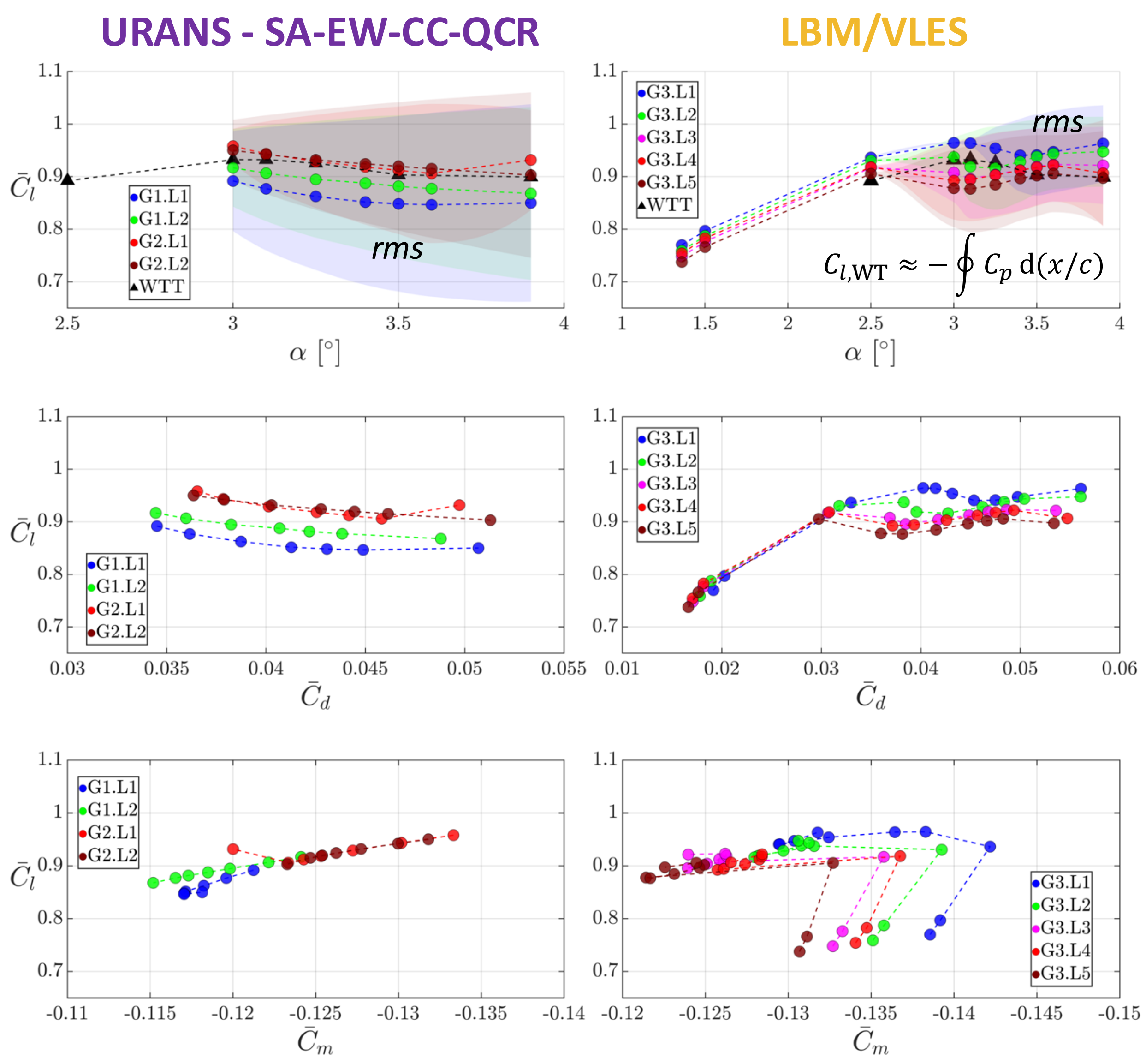
Core scheme (SA-EW-CC-QCR with JST) shows good performance and was therefore utilized in the URANS cases

2nd ≡ MUSCL on

Results – Test Case 1b

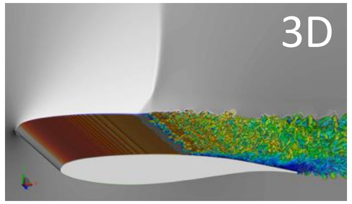
Integrated F&M (URANS vs LBM/VLES)

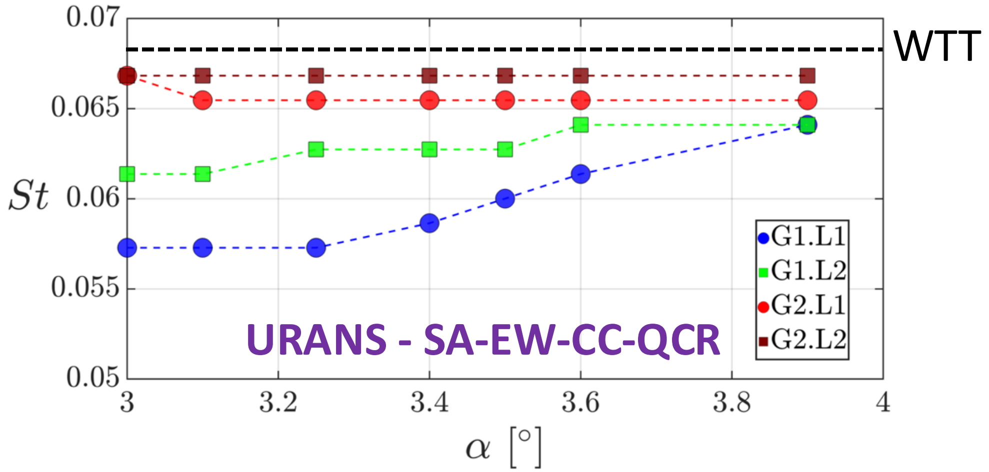
- ❑ **URANS (G1/2.L1/2) vs LBM/VLES (G3.L1-5)**
- ❑ **Lift**
  - ➢ *Less scatter* than RANS
  - ➢ G2 yields closest C_l polars to the WTT data
  - ➢ G1 and G3 show *grid resolution sensitivity*
  - ➢ LBM/VLES results in lower rms C_l than URANS
- ❑ **Drag**
  - ➢ $\alpha = 3.00°$ *scatter*
    - ○ **URANS**: 15 drag counts, $\Delta C_l \approx 0.06$
    - ○ **LBM/VLES**: 30 drag counts, $\Delta C_l \approx 0.03$
  - ➢ $\alpha = 3.90°$ *scatter*
    - ○ **URANS**: 30 drag counts, $\Delta C_l \approx 0.08$
    - ○ **LBM/VLES**: 30 drag counts, $\Delta C_l \approx 0.08$
  - ➢ LBM/VLES results in slightly higher drag at $\alpha \geq 3.00°$

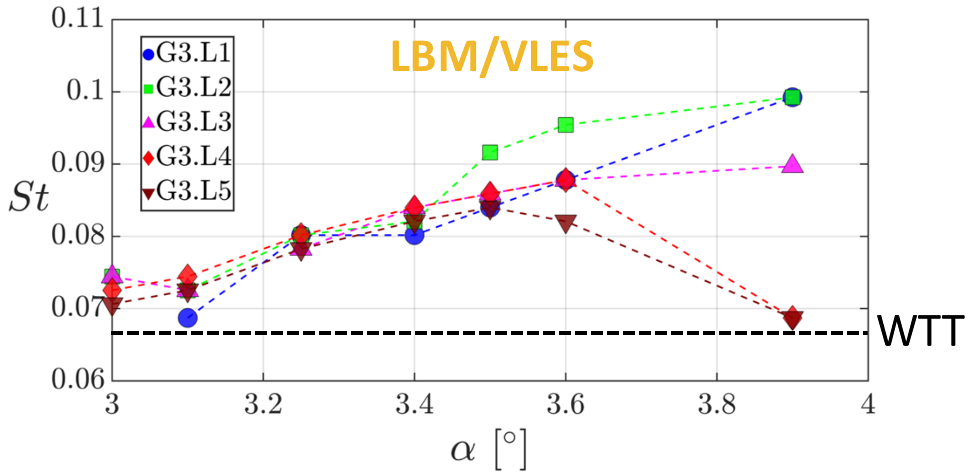


Results – Test Case 1b
Buffet Strouhal Number

- **St vs α (based on C_l signal)**
  - URANS (G1/2.L1/2)
  - LBM/VLES (G3.L1–L5)
  - Experimental benchmark[6]: $St \approx 0.067$, not α-dependent
- URANS with G2 grids shows ***good agreement***
- URANS with G1 grids
  - ***Under-predicting*** the buffet St (15% max error)
  - ***Slight*** α-dependency
- LBM/VLES
  - ***Over-predicting*** the buffet St (~48% max error)
  - ***Strong*** α and grid resolution dependency
  - Preliminary 3D simulations show an improvement to $St \approx 0.073$, suggesting that 2D PowerFLOW simulations may over-predict the flow dynamics (*encouraging*!)





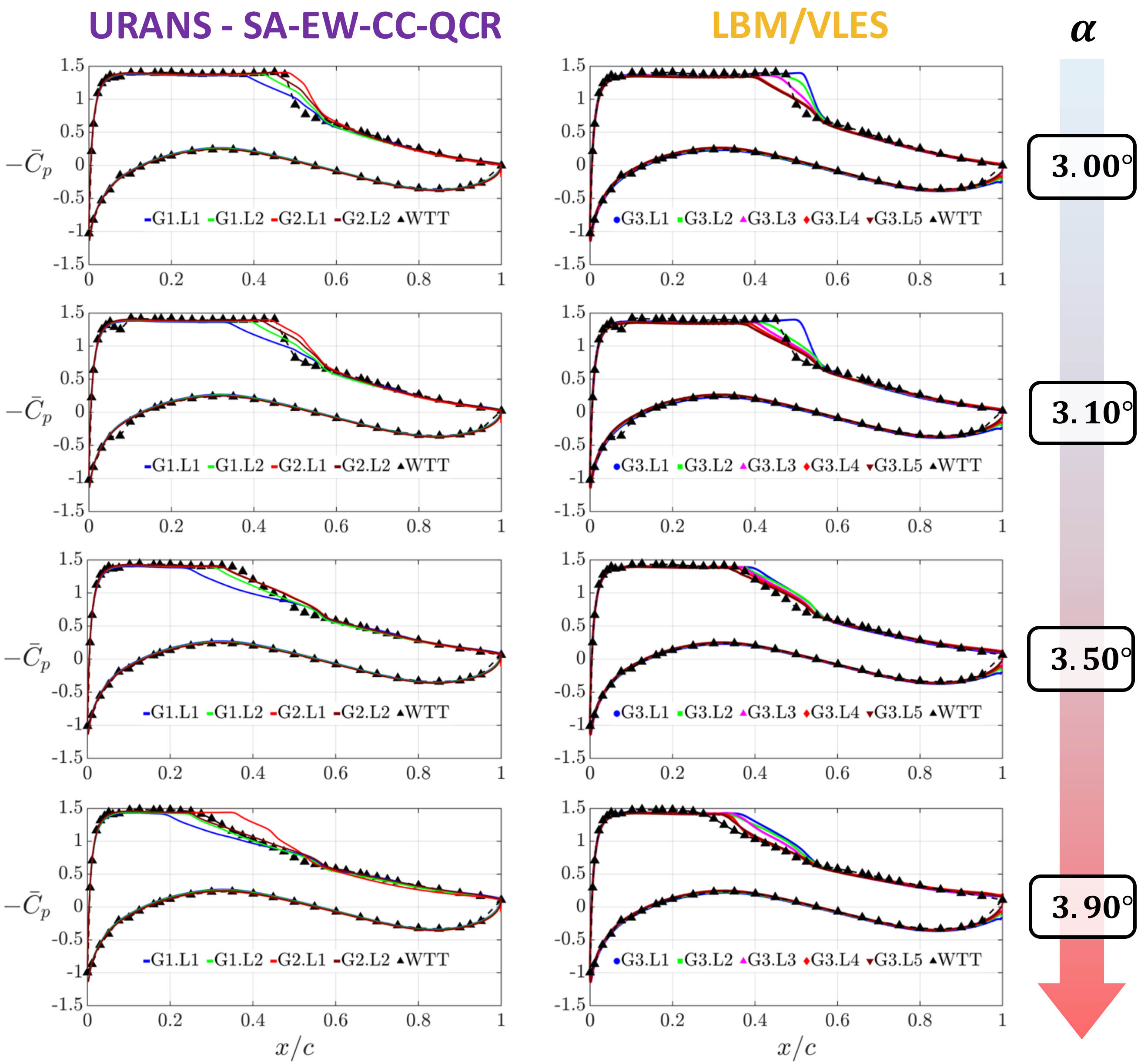


[1] Jacquin L. et al. (2009) "Experimental study of shock oscillation over a transonic supercritical profiles." AIAA Journal 47(9), p. 1985-1994.

Results – Test Case 1b

$\overline{C}_p$ distributions

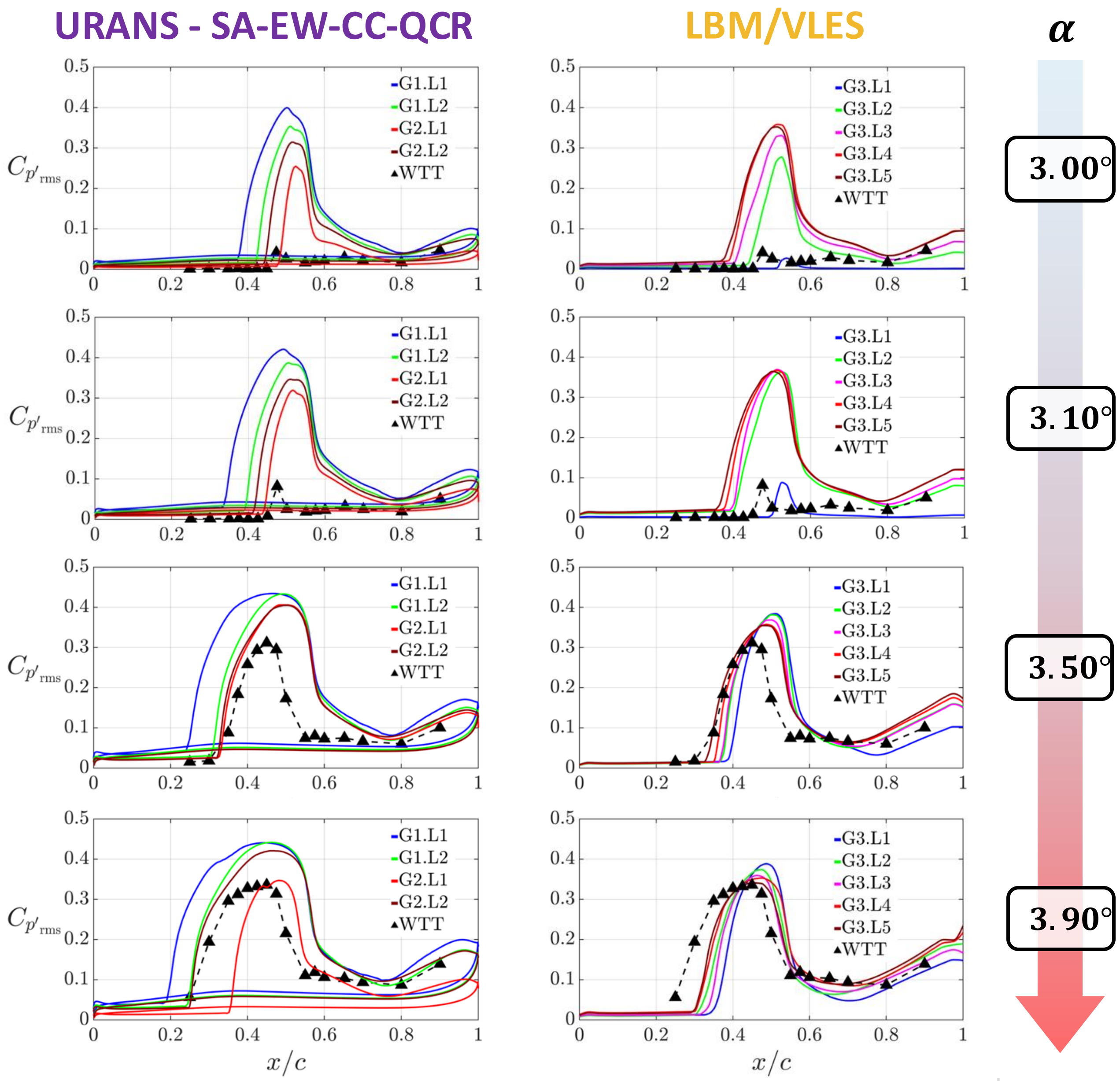
- ❑ **URANS** (G1/2.L1/2) **vs LBM/VLES** (G3.L1-5)
- ❑ *Improved* shock prediction compared to RANS
  - ➢ Smeared pressure gradients at $\alpha \geq 3.10°$ are qualitatively well reproduced
- ❑ **URANS**
  - ➢ *Good* grid convergence
  - ➢ G2.L2 shows the *best agreement* with WTT data
- ❑ **LBM/VLES**
  - ➢ Grid *sensitivity* effects
  - ➢ *Good agreement* with WTT data, with accuracy *similar or slightly improved* than the URANS simulations with G2.L2



Results – Test Case 1b

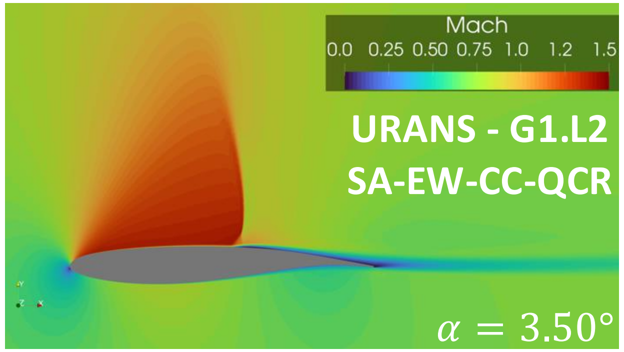
$C_{p',\mathrm{rms}}$ distributions

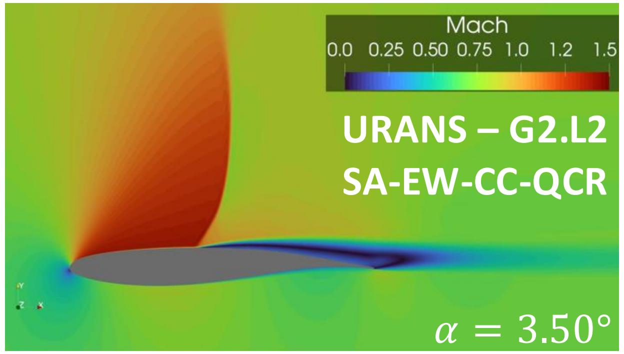
- ❑ **URANS** (G1/2.L1/2) **vs** **LBM/VLES** (G3.L1-5)
  - ➢ LBM/VLES results show slightly *lower* peak $C_{p',\mathrm{rms}}$ values than URANS
  - ➢ Buffet onset is predicted *earlier* with both methods
- ❑ **URANS**
  - ➢ G2 grids show *better agreement* with WTT data than G1 grids
- ❑ **LBM/VLES**
  - ➢ *Good agreement* of LBM/VLES results with WTT data at high α
  - ➢ Good consistency among grid levels G3.L2-L5
  - ➢ G3.L1 shows *surprisingly* good agreement at $\alpha \leq 3.10°$

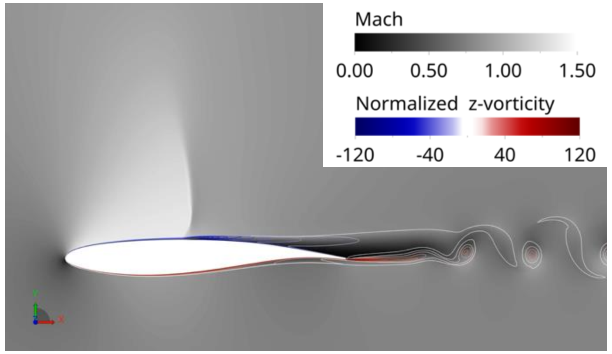


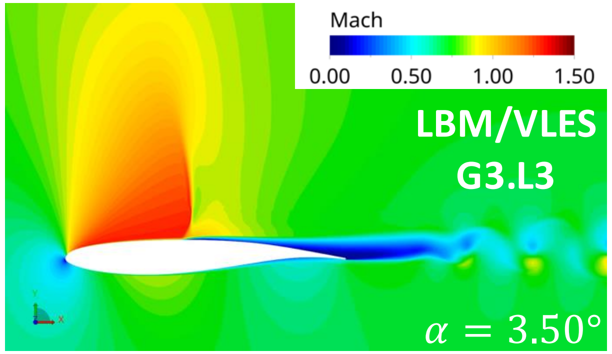
Conclusions

- Both SU2 (URANS) and PowerFLOW (LBM/VLES) can capture the shock buffet phenomenon on the OAT15A
- LBM/VLES approach shows slightly better agreement with WTT data in terms of $\bar{C}_p$, $C_{p',\mathrm{rms}}$ and F&M
- SU2 URANS simulations with structured grids (G2) better capture the buffet frequency
- **Future Work:**
  - Expand to full 3D simulations (Test Case 2)
  - Apply best practices from 2D study









Thank You